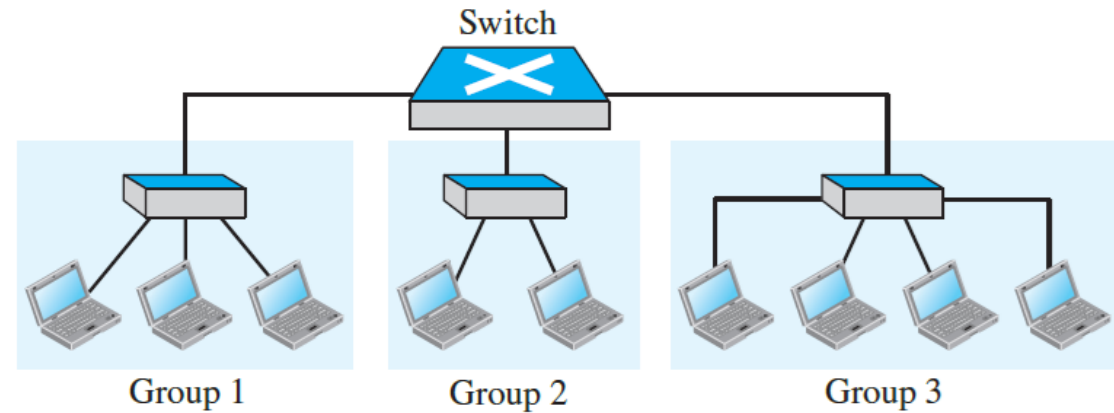
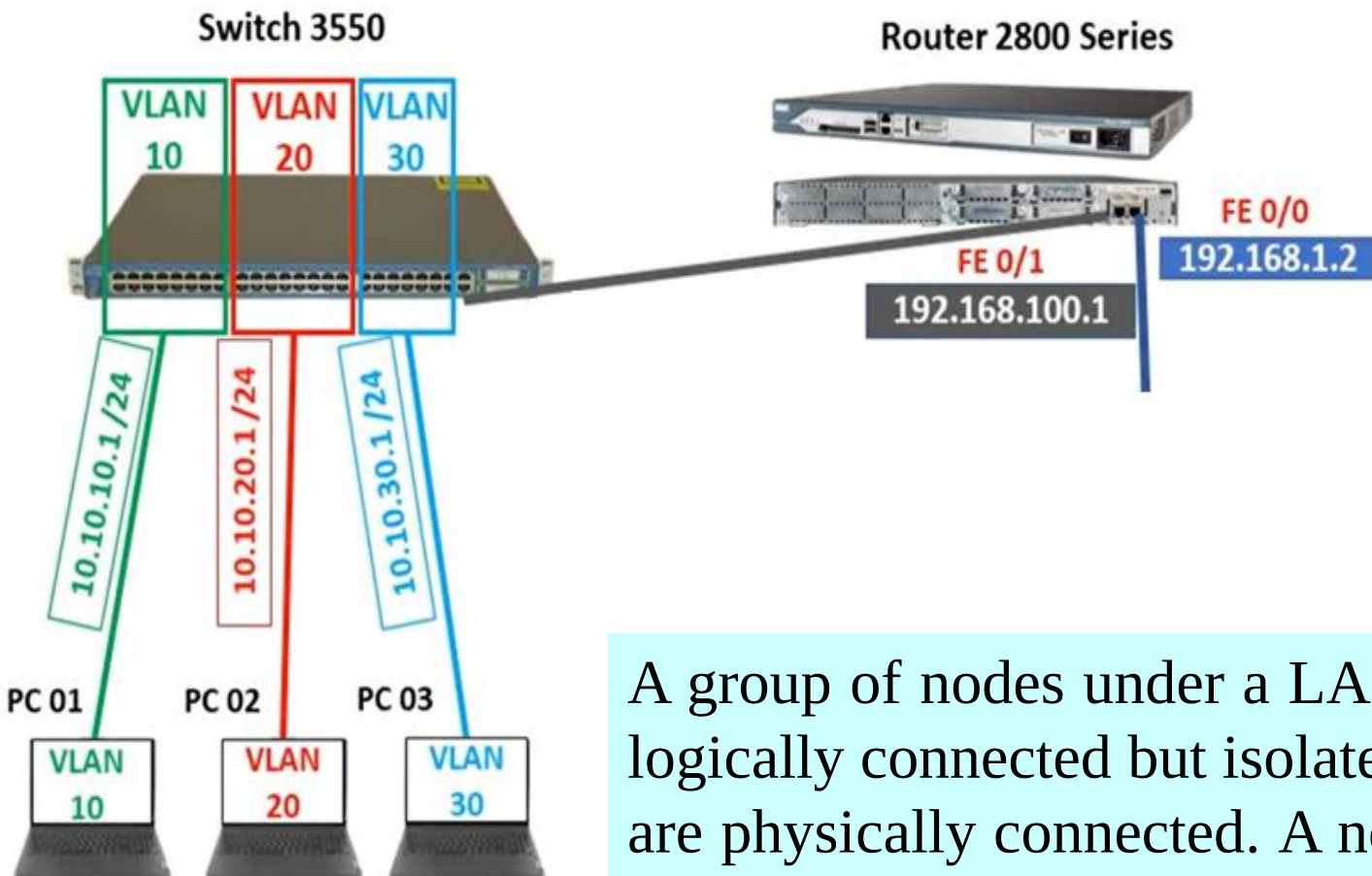
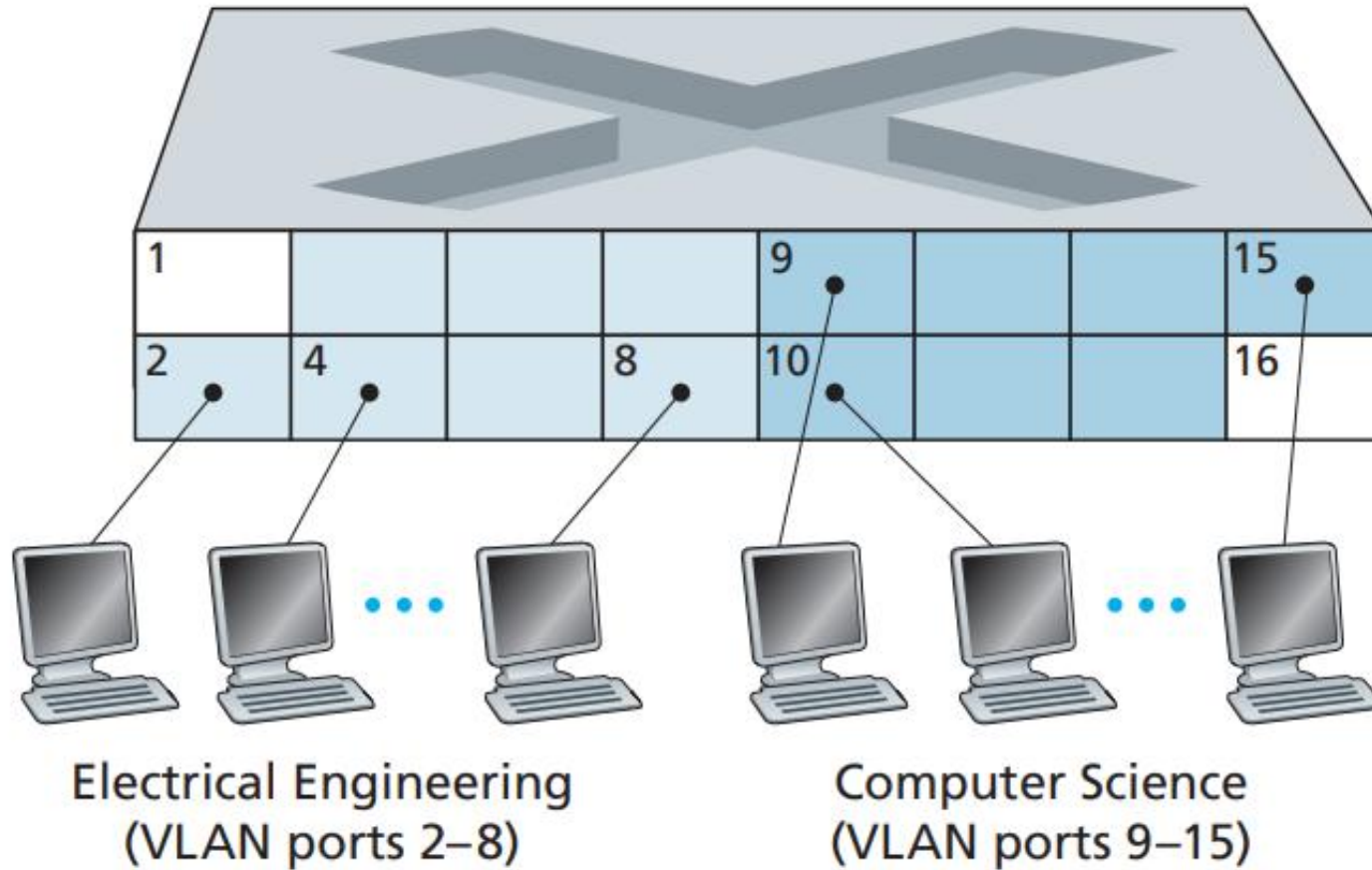


VLAN Configuration with Switch and Router

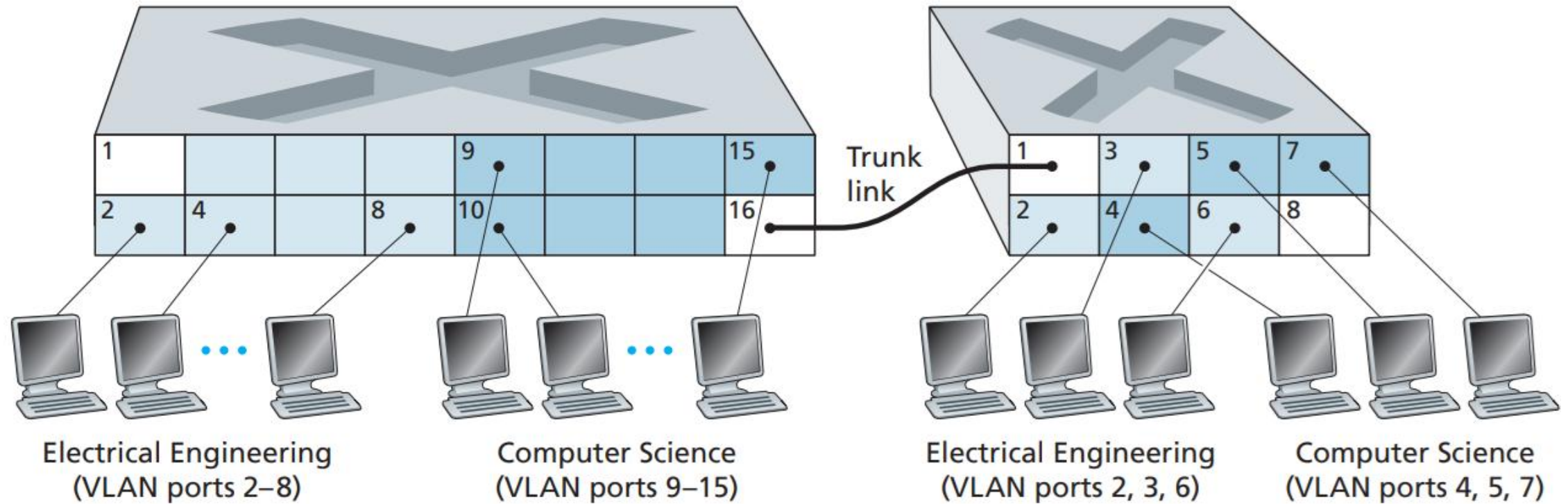


A group of nodes under a LAN (may be under one or more switches) are logically connected but isolated from other groups although all the groups are physically connected. A node is connected with other nodes within the group but not between the groups.



A single switch with two configured VLANs

We can roughly define a virtual local area network (VLAN) as a local area network configured by software, not by physical wiring.



Connecting two VLAN switches with trunk line

	Name	IP of PCs	Interface
VLAN 10	A	192.168.2.1 192.168.2.2	Fa0/1 to Fa0/6
VLAN 20	B	192.168.2.8 192.168.2.12	Fa0/7 to Fa0/14
VLAN 30	C	192.168.2.17 192.168.2.18	Fa0/15 to Fa0/24
VLAN 1		Default	

IP Configuration

IP Configuration

DHCP

Static

IP Address

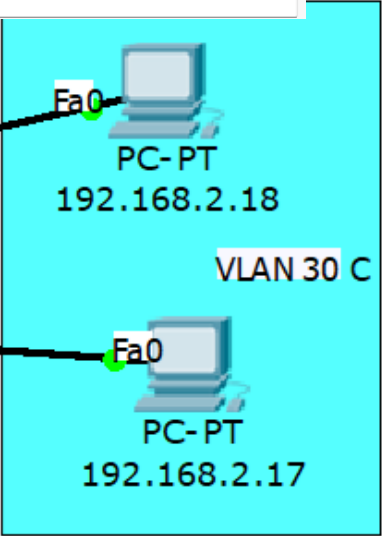
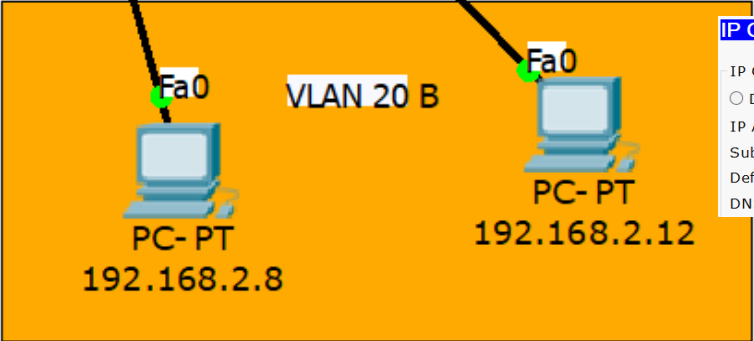
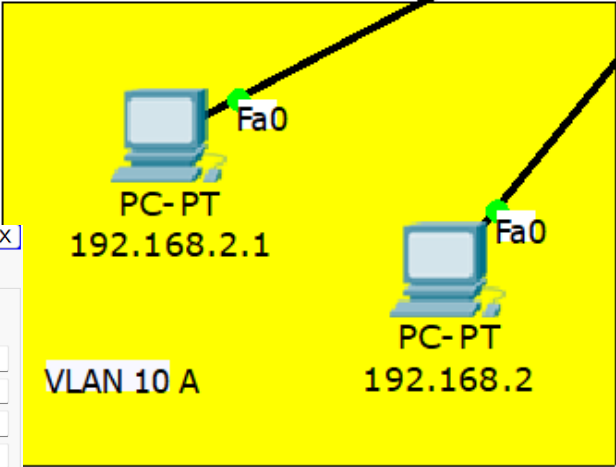
192.168.2.17

Subnet Mask

255.255.255.0

Default Gateway

DNS Server



IP Configuration

IP Configuration

DHCP

Static

IP Address

192.168.2.1

Subnet Mask

255.255.255.0

Default Gateway

DNS Server

IP Configuration

IP Configuration

DHCP

Static

IP Address

192.168.2.8

Subnet Mask

255.255.255.0

Default Gateway

DNS Server

```
Switch>en
```

```
Switch#conf t
```

```
Switch(config)#vlan 10
```

```
Switch(config-vlan)#name A
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#vlan 20
```

```
Switch(config-vlan)#name B
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#vlan 30
```

```
Switch(config-vlan)#name C
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#int range fa0/1-6
```

```
Switch(config-if-range)#switchport access vlan 10
```

```
Switch(config-if-range)#exit
```

```
Switch(config)#int range fa0/7-14
```

```
Switch(config-if-range)#switchport access vlan 20
```

```
Switch(config-if-range)#exit
```

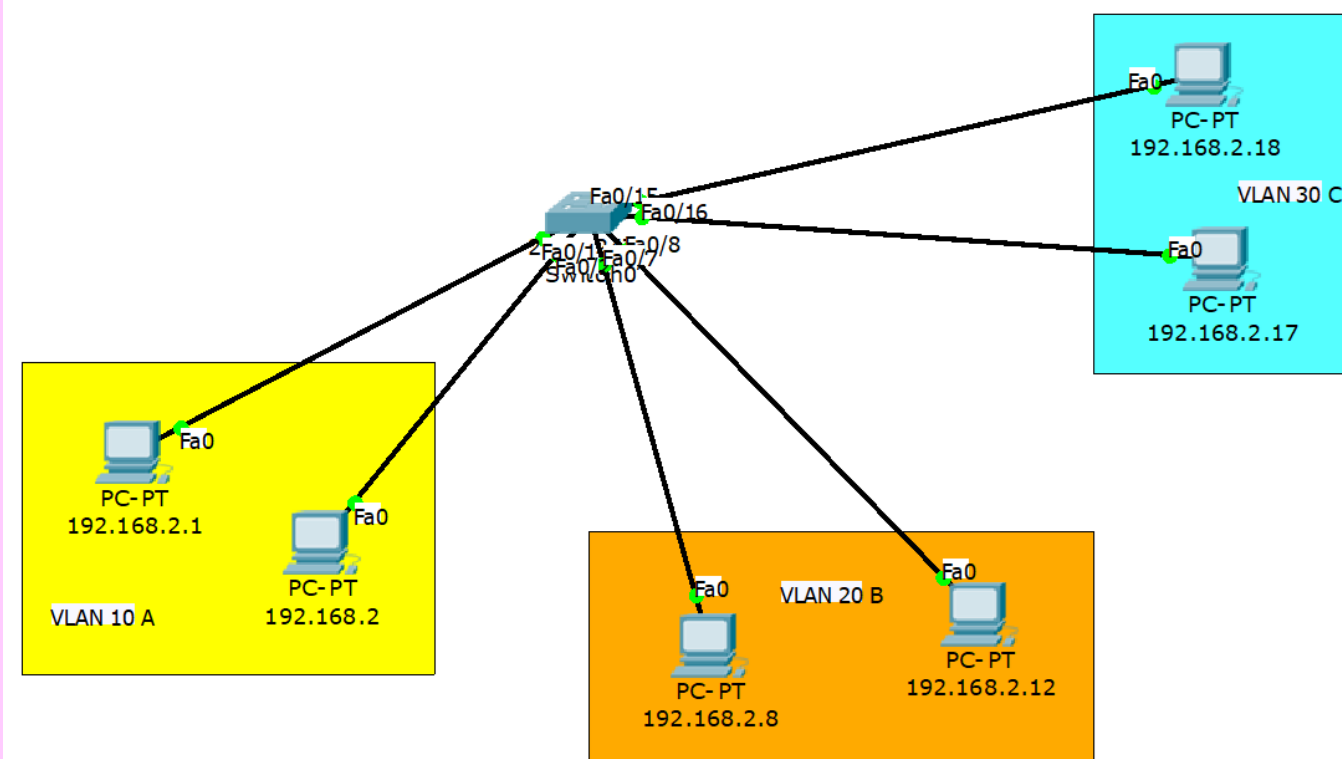
```
Switch(config)#int range fa0/15-24
```

```
Switch(config-if-range)#switchport access vlan 30
```

```
Switch(config-if-range)#end
```

```
Switch#sh vlan brief
```

	Name	IP of PCs	Interface
VLAN 10	A	192.168.2.1 192.168.2.2	Fa0/1 to Fa0/6
VLAN 20	B	192.168.2.8 192.168.2.12	Fa0/7 to Fa0/14
VLAN 30	C	192.168.2.17 192.168.2.18	Fa0/15 to Fa0/24



Switch#sh vlan brief

VLAN Name	Status	Ports
-----------	--------	-------

1	default	active	Gig1/1, Gig1/2
10	A	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6
20	B	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14
30	C	active	Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

Now verify using ping and ICMP

VLAN under sub-interface

IP Configuration

IP Configuration

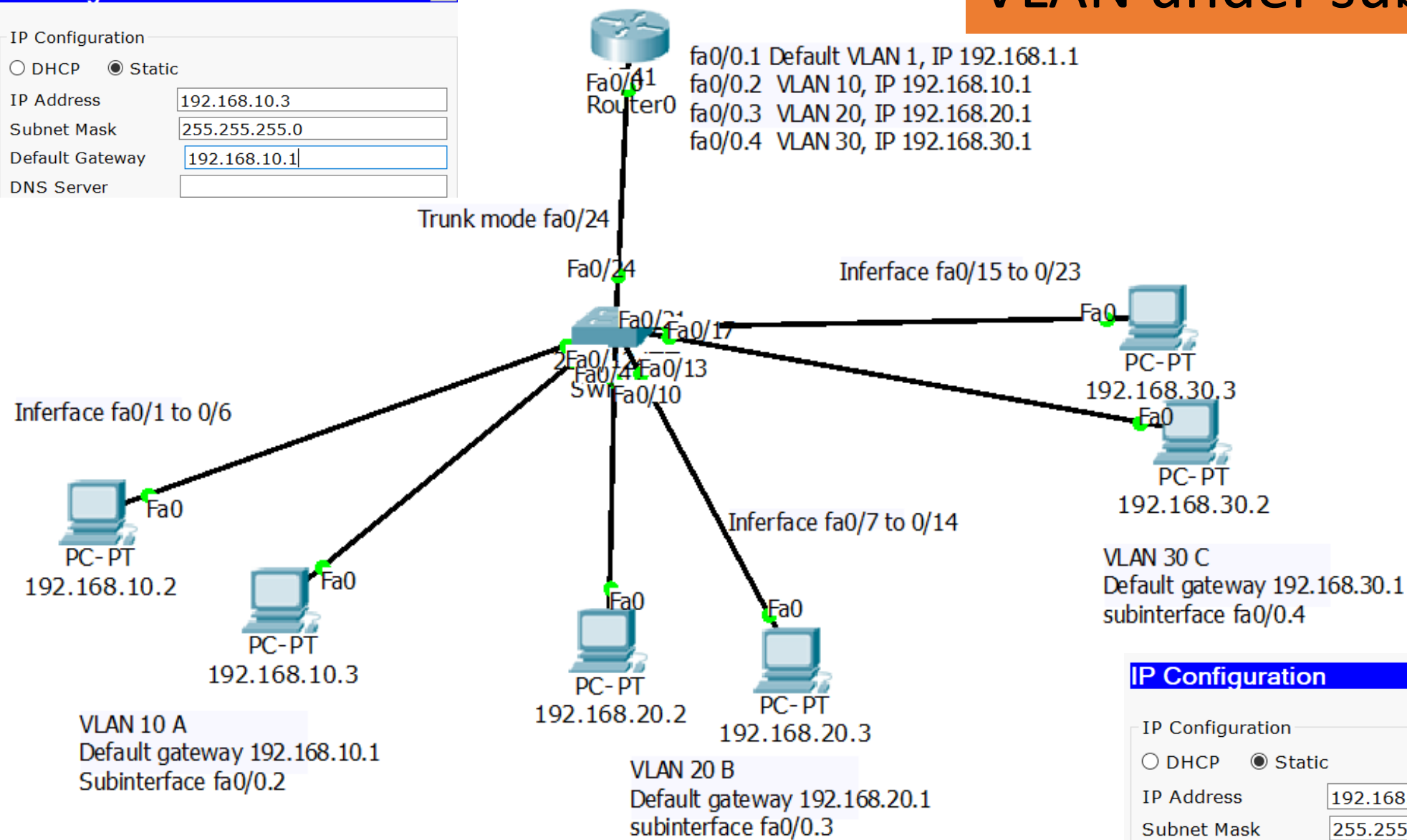
☐ DHCP ☒ Static

IP Address 192.168.10.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server



IP Configuration

IP Configuration

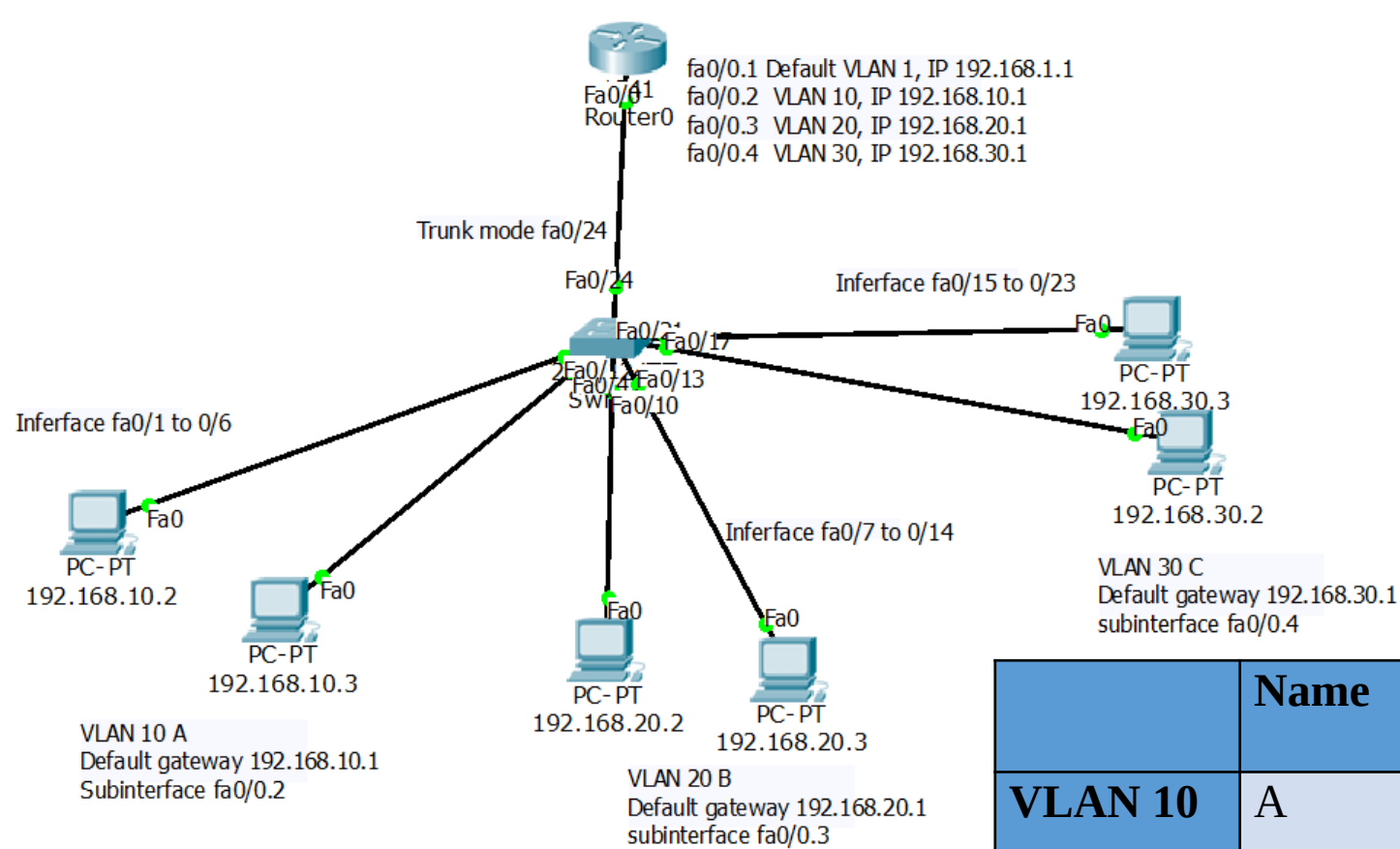
☐ DHCP ☒ Static

IP Address 192.168.30.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.30.1

DNS Server



	Name	IP of PCs	Default Gateway	Interface
VLAN 10	A	192.168.10.2 192.168.10.3	192.168.10.1 Sub interface fa0/0.2	Fa0/1 to Fa0/6
VLAN 20	B	192.168.20.2 192.168.20.3	192.168.20.1 Sub interface fa0/0.3	Fa0/7 to Fa0/14
VLAN 30	C	192.168.30.2 192.168.30.3	192.168.30.1 Sub interface fa0/0.4	Fa0/15 to Fa0/23
VLAN 1	Default VLAN Router itself		192.168.1.1 Sub interface fa0/0.1	The interface of the router Fa0/0

Switch>en

Switch#conf t

Switch(config)#vlan 10

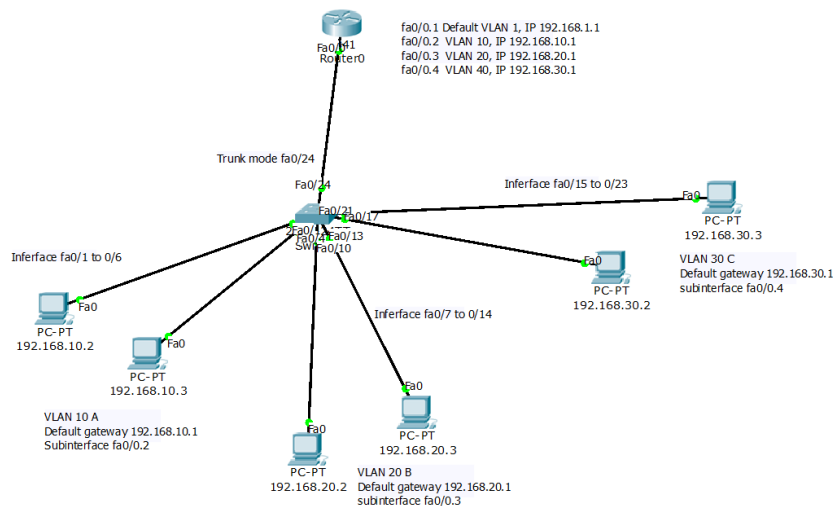
Switch(config-vlan)#name A

Switch(config-vlan)#exit

Switch(config)#vlan 20

Switch(config-vlan)#name B

Switch(config-vlan)#exit



Switch(config)#vlan 30

Switch(config-vlan)#name C

Switch(config-vlan)#exit

Switch(config)#int range fa0/1-6

Switch(config-if-range)#switchport access vlan 10

Switch(config-if-range)#exit

Switch(config)#int range fa0/7-14

Switch(config-if-range)#switchport access vlan 20

Switch(config-if-range)#exit

Switch(config)#int range fa0/15-23

Switch(config-if-range)#switchport access vlan 30

Switch(config-if-range)#exit

Switch(config)#int fa0/24

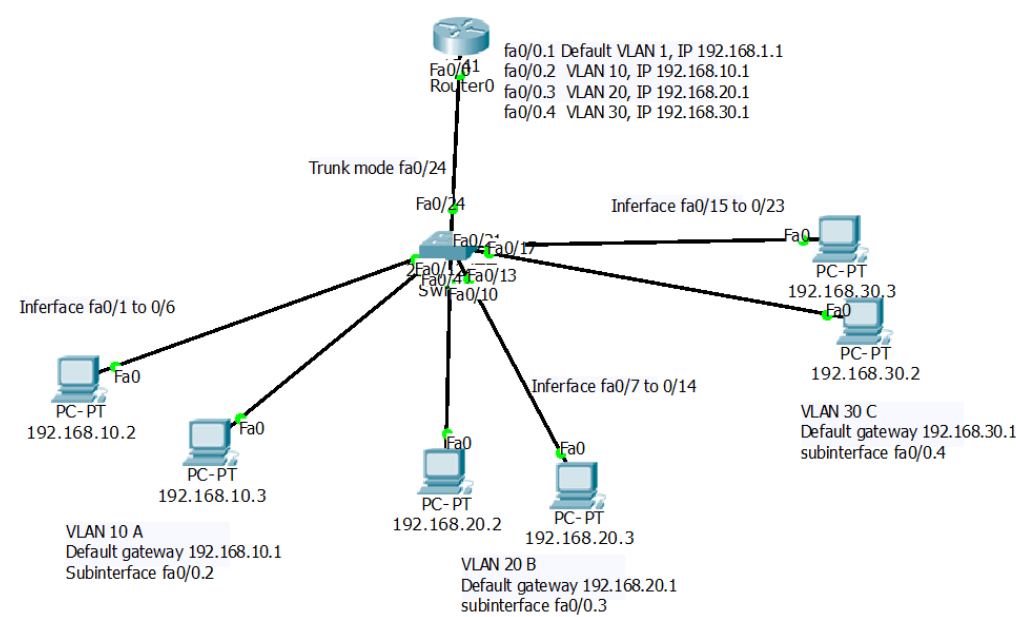
Switch(config-if)#switchport mode trunk

Switch(config-if)#end

VLAN Name	Status	Ports

1 default	active	Fa0/24, Gig1/1, Gig1/2
10 A	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6
20 B	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 ,Fa0/11, Fa0/12, Fa0/13, fa0/14
30 C	active	Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Router>en
Router#conf t
Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#int fa0/0.1
Router(config-subif)#encapsulation dot1q 1
Router(config-subif)#ip add 192.168.1.1 255.255.255.0
Router(config-subif)#int fa0/0.2
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip add 192.168.10.1 255.255.255.0
Router(config-subif)#int fa0/0.3
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip add 192.168.20.1 255.255.255.0
Router(config-subif)#int fa0/0.4
Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip add 192.168.30.1 255.255.255.0
Router(config-subif)#end
```



Router#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

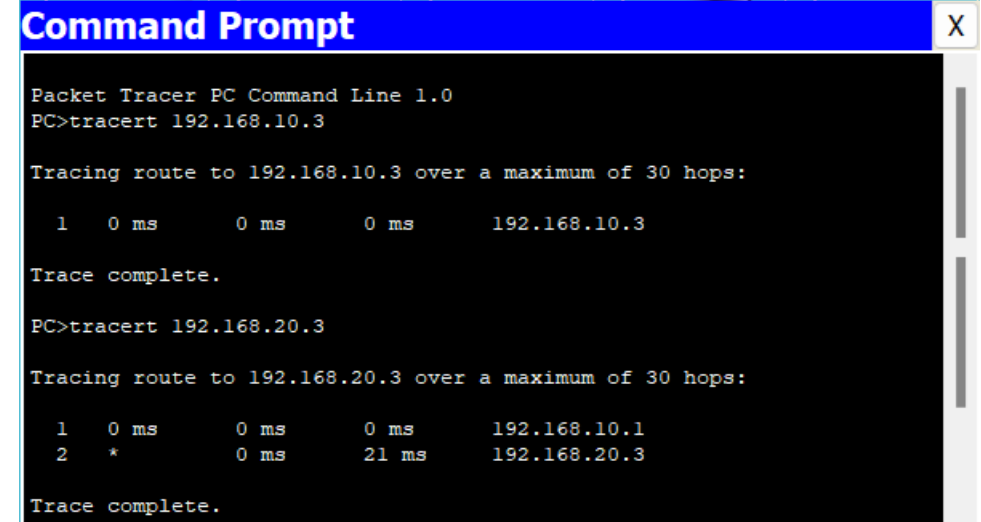
Gateway of last resort is not set

C 192.168.1.0/24 is directly connected, FastEthernet0/0.1

C 192.168.10.0/24 is directly connected, FastEthernet0/0.2

C 192.168.20.0/24 is directly connected, FastEthernet0/0.3

C 192.168.30.0/24 is directly connected, FastEthernet0/0.4



```
Command Prompt
Packet Tracer PC Command Line 1.0
PC>tracert 192.168.10.3

Tracing route to 192.168.10.3 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.10.3

Trace complete.

PC>tracert 192.168.20.3

Tracing route to 192.168.20.3 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.10.1
  2  *         0 ms      21 ms     192.168.20.3

Trace complete.
```

Verify the network using tracert, ping and ICMP